

CSE422 Lab Project Report

Heart Disease Prediction using Machine Learning

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1. Introduction

Heart disease is one of the major causes of death worldwide and early detection can play a very important role in saving lives. With the help of machine learning, it is possible to analyze patient data and predict whether someone is likely to have heart disease or not.

The main goal of this project is to build a machine learning model that can predict heart disease based on different medical attributes such as age, cholesterol level, heart rate and chest pain type.

In this project, we go through the full pipeline starting from understanding the dataset, cleaning and preprocessing the data, training different models, and finally comparing their performance.

The motivation behind this work is to understand how machine learning techniques can be applied in real-world healthcare problems.

2. Dataset Description

The dataset used in this project contains 920 data points and 16 features, including one target variable called “num”.

Problem Type

This is a classification problem because the target variable represents categories. Initially, the dataset had multiple classes (0–4), but for simplicity, it was converted into a binary format:

- 0 → No heart disease
- 1 → Presence of heart disease

Feature Types

The dataset includes both:

- Numerical features (age, cholesterol, blood pressure)
- Categorical features (chest pain type, thal, slope)

Since machine learning models cannot directly work with categorical data, these features need to be converted into numerical form through encoding.

Correlation Analysis

A heatmap was used to understand how different features are related to each other and to the target variable.

From the heatmap, it can be seen that:

- Some features like “oldpeak” and “thalch” show a noticeable relationship with the target
- A few features are also correlated with each other
- However, no single feature strongly determines the outcome

This suggests that heart disease prediction depends on a combination of multiple features rather than just one.

Imbalanced Dataset

At the beginning, the dataset was imbalanced, meaning some classes had many more samples than others.

After converting the target into binary form, the dataset became almost balanced, with a similar number of samples in both classes.

This makes the models more reliable and reduces bias toward one class.

Exploratory Data Analysis (EDA)

Some basic visualizations were created to better understand the data.

- Age vs Disease: Older individuals seem to have a higher chance of heart disease
- Chest Pain Type: Certain types of chest pain are more common in patients with disease
- Max Heart Rate (thalch): Lower values are often associated with heart disease
- Oldpeak: Higher values are more common among patients with disease

Overall, these observations show that several features contribute together in predicting heart disease.

3. Dataset Pre-processing

Handling Missing Values

The dataset had missing values in multiple columns. Simply removing those rows would result in losing a lot of data, so different imputation techniques were used instead.

- Numerical features were filled using the mean, since they are continuous
- The “ca” feature was filled using the median, as it is less sensitive to outliers
- Categorical features were filled using the most frequent value

Although some features like “thal” had a large number of missing values, using mode imputation helped keep the dataset size intact, even though it might introduce a bit of bias.

Handling Categorical Values

Several features were categorical, and machine learning models cannot process them directly.

To solve this, Label Encoding was applied to convert all categorical values into numeric form. This step is necessary to make the dataset suitable for model training.

Feature Scaling

The features in the dataset had very different ranges. For example, cholesterol values can be in the hundreds, while other features are much smaller.

To handle this, StandardScaler was used to normalize the data so that:

- Mean becomes 0
- Standard deviation becomes 1

This step is especially important for models like KNN and Neural Networks.

4. Dataset Splitting

The dataset was divided into:

- 80% for training
- 20% for testing

Stratified sampling was used to ensure that both training and testing sets have a similar distribution of classes.

This helps in getting a more reliable evaluation of model performance.

5. Model Training & Testing

Three supervised models and one unsupervised method were applied.

K-Nearest Neighbors (KNN)

KNN was used as a distance based model. Different values of k were tested, and the best one was selected based on accuracy.

Since KNN relies on distance, feature scaling played an important role here.

Logistic Regression

Logistic Regression was used as a simple and interpretable model for classification.

It performs well for binary classification problems and serves as a good baseline.

Neural Network (MLP)

A Neural Network with two hidden layers was used to capture more complex patterns in the data.

Compared to other models, it can learn non-linear relationships more effectively.

K-Means Clustering

The dataset was also treated as an unsupervised problem using K-Means clustering.

Two clusters were created, and PCA was used to visualize the clusters in 2D space. The silhouette score was used to evaluate clustering quality.

6. Model Comparison

After training all the models, their performance was compared using accuracy, precision, recall, confusion matrices and ROC curves. This helped understand not just which model performs best, but also how they make mistakes.

K-Nearest Neighbors (KNN)

KNN turned out to be the best-performing model, achieving an accuracy of 84.78%.

- Precision (Class 1): 0.89
- Recall (Class 1): 0.83

The model is quite good at identifying patients with heart disease. It makes fewer false positive errors and maintains a good balance between precision and recall.

KNN relies on distance between data points and since the data was properly scaled, it performed effectively. Also, choosing the optimal value of $k = 13$ helped reduce noise and improve prediction stability.

Logistic Regression

Logistic Regression achieved an accuracy of 81.52%, which is slightly lower than KNN but still quite solid.

- Precision (Class 1): 0.83
- Recall (Class 1): 0.84

This model is very balanced. It doesn't heavily favor one class over the other and gives consistent results.

Logistic Regression assumes a linear relationship between features and the target. Since the dataset has some linear patterns, it performs reasonably well, but it struggles with more complex relationships.

Neural Network (MLP)

The Neural Network gave the lowest accuracy of 77.17%.

- Precision (Class 1): 0.79

- Recall (Class 1): 0.80

The model is still doing okay, but it makes more mistakes compared to the other models. Both false positives and false negatives are higher.

Neural Networks usually need a larger dataset to perform well. In this case, the dataset is relatively small, so the model doesn't get enough data to learn complex patterns properly. Also, it may require more tuning.

Confusion Matrix (Overall View)

Looking at the confusion matrices:

- Most predictions fall into the correct category
- KNN has the least number of mistakes
- Neural Network shows more misclassification

So overall, KNN is the most reliable among the three.

Precision & Recall Comparison

- KNN → Best precision (fewer false alarms)
- Logistic Regression → Most balanced
- Neural Network → Slightly weaker

This shows that even though all models are decent, KNN handles the classification task more effectively.

ROC Curve & AUC

From the ROC curves:

- KNN shows strong class separation
- Logistic Regression is stable and predictable
- Neural Network is comparatively weaker

This again supports the earlier results.

K-Means Clustering (Unsupervised Learning)

- Silhouette Score: 0.138

This is a very low score, which indicates that the clusters are not well separated.

The dataset does not naturally form clear groups. This suggests that clustering is not very suitable for this problem, and supervised learning works better here.

Overall Comparison

- KNN → Best performance
- Logistic Regression → Most stable
- Neural Network → Lowest performance

7. Conclusion

This project shows that machine learning can be used to predict heart disease with a fairly good level of accuracy.

Among all the models:

- KNN performed the best
- Logistic Regression gave consistent results
- Neural Network did not perform as well as expected

KNN worked well because it relies on similarity between data points and the dataset was properly scaled. Logistic Regression performed nicely because there are some linear patterns in the data.

On the other hand, the Neural Network didn't perform as well, mainly because the dataset is not large enough for it to fully learn complex patterns.

Challenges faced

During the project, a few challenges came up:

- Handling missing values without losing too much data
- Encoding categorical variables correctly
- Making sure features were scaled properly
- Choosing the right models and parameters

Final thoughts

One important takeaway from this project is that more complex models are not always better. In this case, simpler models like KNN and Logistic Regression actually performed better than the Neural Network.

It also shows how important preprocessing is. Small steps like handling missing values and scaling features had a big impact on the final results.